

HW05 Tasks 1-3 - Performance Test Design, Execution, AI Review, and Continuous Testing

Student ID: 23127035
Execution date: 2026-09-01 (Asia/Ho_Chi_Minh)
SUT: EShop backend from `../hw04/eshop-sut-main`
Selected workflow: WF-04 - Admin Login -> Import Product CSV -> Search Imported Product -> View Product Detail
Tool: Apache JMeter 5.6.3, non-GUI execution

1. Scope and workflow correction

The original FR03-FR09-FR16 chain joined unrelated password-reset, coupon-validation, and product-import operations. After the student supplied the team's three selected workflows (WF-01 through WF-03), it was replaced with the distinct WF-04 admin product-publishing journey. WF-04 has one coherent goal: an administrator authenticates, imports a new product, finds it in the public catalogue, and opens its detail record.

The performance evidence in Tasks 1 and 2 uses only the accepted WF-04 results; Task 3 builds its continuous-testing proposal on that candidate baseline. The superseded workflow remains recoverable from Git history and an ignored local archive, but it is not mixed into the submission baseline. Against the three team workflows explicitly supplied by the student, WF-04 is not duplicated; this does not claim knowledge of any allocation outside that list.

No screenshot, voice recording, YouTube URL, GitHub Issue, or human-approval statement was generated by AI. Student-owned evidence remains listed in `STUDENT_EVIDENCE_REQUIRED.md`.

2. Test environment and isolation

Component	Actual value
Hostname	<code>MacBook-Air-cua-KunDa.local</code>
Hardware	MacBook Air <code>Mac16,12</code> , Apple M4, 10 cores, 16 GB RAM, arm64
Operating system	macOS 26.5.1 (25F80), Darwin 25.5.0
Backend	Node.js v26.4.0, Express 5.2.1, SQLite 3.51.0
JMeter	5.6.3, non-GUI mode
Measurement endpoint	<code>http://127.0.0.1:3001</code>
Isolation	Temporary copy of unchanged HW04 backend source with only the listening port changed; separate SQLite file for every run

Every accepted run used the isolated runtime on port 3001. This prevents unrelated browser/frontend traffic on the default port from contaminating product counts or throughput.

3. Coherent end-to-end workflow

Order	Feature/group	Request	Correlation and business assertion
1	FR02 / auth-heavy	<code>POST /api/login</code>	HTTP 200; extract JWT, user ID, and role; require a non-empty token and <code>role=admin</code> .

Order	Feature/group	Request	Correlation and business assertion
2	FR16 / transactional write	<code>POST /api/admin/import-products</code>	Send the correlated Bearer token and a runtime-unique product; require <code>inserted=1</code> , empty errors, and <code>1/1</code> in the message.
3	FR05 / read-heavy	<code>GET /api/products?search=...</code>	Search the exact unique name; require exactly one match and correlate its product ID.
4	FR06 / read-heavy	<code>GET /api/products/{product_id}</code>	Use the correlated ID; require the same ID, name, and price as the imported product.

The final detail lookup proves that the product created by FR16 is available through the catalogue read path. FR16 is write-heavy but the source does not wrap a batch in an explicit SQLite transaction, so it is not falsely described as atomic.

4. Data, correlation, and assertions

`data/performance_users.csv` has 80 synthetic admin accounts and the fields `email,password,product_prefix,product_price,product_description,product_image_url,category_id`. This exceeds peak concurrency of 30 VUs. Each iteration creates a unique product name from the CSV prefix, thread number, iteration number, and current millisecond.

CSV sharing is `All threads`, recycling is explicit, and one account is never concurrently shared at the tested peak. The JWT and product ID are extracted dynamically. Every sampler has an HTTP 200 assertion plus a business assertion; an HTTP response cannot pass merely because the socket completed. The shared 200-500 ms think-time timer is placed at Test Plan scope and therefore applies to all four actions in every thread group.

5. Workload models

Plan	Workload	Listener/report view	Purpose
<code>23127035_Load_20260901.jmx</code>	6 VUs, 15 s ramp, 120 s	Summary Report	Small stable interactive load after Smoke.
<code>23127035_Stress_20260901.jmx</code>	6, 12, 24 VUs; 10 s ramp and 60 s per stage	Aggregate Report	Doubles concurrency and isolates degradation by stage.
<code>23127035_Spike_20260901.jmx</code>	3 VUs/45 s, 30 VUs/30 s, 3 VUs/45 s recovery	View Results Tree, disabled during run	Abrupt tenfold burst followed by recovery.
<code>23127035_Soak_20260901.jmx</code>	30 VUs, 30 s ramp, 410 iterations/thread	HTML dashboard/raw JTL	Completes 12,300 workflows in about ten minutes without relying on an unstable wall-clock scheduler.

The Soak workload was retained at 30 VUs only after the real 30-VU Spike burst passed. Stop/invalidity conditions were backend exit, JMeter non-zero exit, any business assertion failure, missing raw/report evidence, or database state inconsistent with successful imports.

6. Execution, state control, and excluded attempt

Before every run the isolated backend recreated the canonical 2 users, 5 products, and 4 coupons. The runner required those counts to remain stable for three checks, inserted and verified exactly 80 admin accounts, captured pre-state, and sampled the exact Node PID once per second. Each accepted run therefore started with 82 users, 81 admins, 5 products, 0 orders, and 0 coupon-usage rows.

Smoke completed all four samplers with 0 failures and changed products from 5 to 6. Load, Stress, and Spike then ran from independent clean database states. The first WF-04 Soak attempt is excluded: all 24,528 samples passed, but JTL traffic covered only 305.20 seconds while the wall clock jumped forward and caused the 600-second scheduler to stop early. Full evidence is retained in [evidence/inconclusive/23127035_Soak_20260831_wf04_clock_jump](#). The accepted rerun used 410 iterations per thread, completed exactly 12,300 workflows, and spanned 618.94 seconds in the JTL.

7. Accepted results

Percentiles use nearest rank over JMeter `elapsed`. `Samples/s` counts HTTP samplers. The exact workflow count is the number of successful final FR06 detail samplers; it is not inferred by dividing total samples.

Scenario	Samples	Failed	Error %	Mean ms	p90 ms	p95 ms	p99 ms	Samples/s	Exact workflows	Workflows/s	Peak CPU %	Peak RSS MB
Load	1,912	0	0.00	2.35	4	5	6	16.04	477	4.00	6.9	72.59
Stress	6,514	0	0.00	1.63	3	4	5	36.33	1,613	9.00	13.7	114.48
Spike	3,216	0	0.00	1.76	3	4	5	26.98	793	6.65	15.4	101.77
Soak	49,200	0	0.00	2.29	3	4	6	79.49	12,300	19.87	22.3	136.67

Stage-level evidence

Stage	Samples	Failed	p95 ms	Samples/s
Stress 6 VUs	932	0	5	15.74
Stress 12 VUs	1,873	0	4	31.44
Stress 24 VUs	3,709	0	4	62.21
Spike baseline 3 VUs	360	0	5	8.13
Spike burst 30 VUs	2,482	0	4	84.03
Spike recovery 3 VUs	374	0	4	8.38

Stress throughput scaled with concurrency while p95 improved from 5 to 4 ms and no assertion failed. Spike recovery returned to 8.38 samples/s and p95 4 ms, comparable to the 8.13 samples/s and p95 5 ms baseline.

8. Database and resource consistency

Scenario	Products before	Successful FR16 imports	Products after	Consistent?
Load	5	477	482	Yes
Stress	5	1,634	1,639	Yes
Spike	5	807	812	Yes
Soak	5	12,300	12,305	Yes

Resource figures are for the exact backend PID. macOS process CPU is not normalized across ten cores. RSS is resident memory; VSZ is not treated as physical usage. Soak RSS started at 67.64 MB, peaked at 136.67 MB, and ended at 51.86 MB, so the peak is not called a memory ceiling or a leak.

9. Endurance conclusion

The highest stable load observed was **30 VUs for 618.94 seconds**, completing **49,200 HTTP samples and 12,300 workflows** at **79.49 samples/s and 19.87 workflows/s**. Overall p95 was **4 ms**, p99 **6 ms**, error rate **0.00%**, peak Node CPU **22.3%**, and peak RSS **136.67 MB**.

The maximum elapsed value was 1,151 ms. It occurred in a brief cluster affecting several sampler types, while p99 remained 6 ms; it is retained as an outlier rather than substituted for the distribution. No tested stage reached failure or saturation, so 30 VUs is a lower-bound stable envelope, not absolute capacity.

10. AI review and corrections

Problem found	Correction	Evidence/reason
The first workflow connected unrelated FR03, FR09, and FR16 actions.	Replaced it with WF-04 admin product publishing.	One business goal now links authentication, write, search, correlation, and detail verification; it is distinct from the three team workflows supplied.
Sample throughput could be labelled workflow throughput.	Count successful final FR06 samplers as exact completions.	Soak is 79.49 samples/s but 19.87 workflows/s.
Whole Stress and Spike averages could be compared as peak levels.	Compare their stage windows.	Stress 24 VUs is 62.21 samples/s; Spike burst 30 VUs is 84.03 samples/s.
A 1,151 ms maximum could be called a tail failure.	Report max separately from p95 4 ms and p99 6 ms.	One short outlier cluster does not describe 49,200 samples.
Duration-based Soak was assumed robust to host clock changes.	Excluded the short attempt and used fixed completed iterations.	Accepted JTL spans 618.94 seconds and contains exactly 12,300 final samplers.

These are AI-prepared technical corrections. They become the assignment's mandatory human review only when the student checks the raw evidence and explicitly approves or amends them.

11. Task 2 threshold proposal

The following are proposed same-environment regression gates, not an SLA: fail any business assertion; warn/fail when p95 exceeds 7.5/10 ms or p99 exceeds 9/12 ms; fail below 14.44 samples/s at Load 6 VUs, 55.99 at Stress 24 VUs, 75.63 during the Spike burst, or 71.54 samples/s and 17.89 workflows/s for Soak. Investigate Node CPU above 30% or RSS above 165 MB without automatically failing.

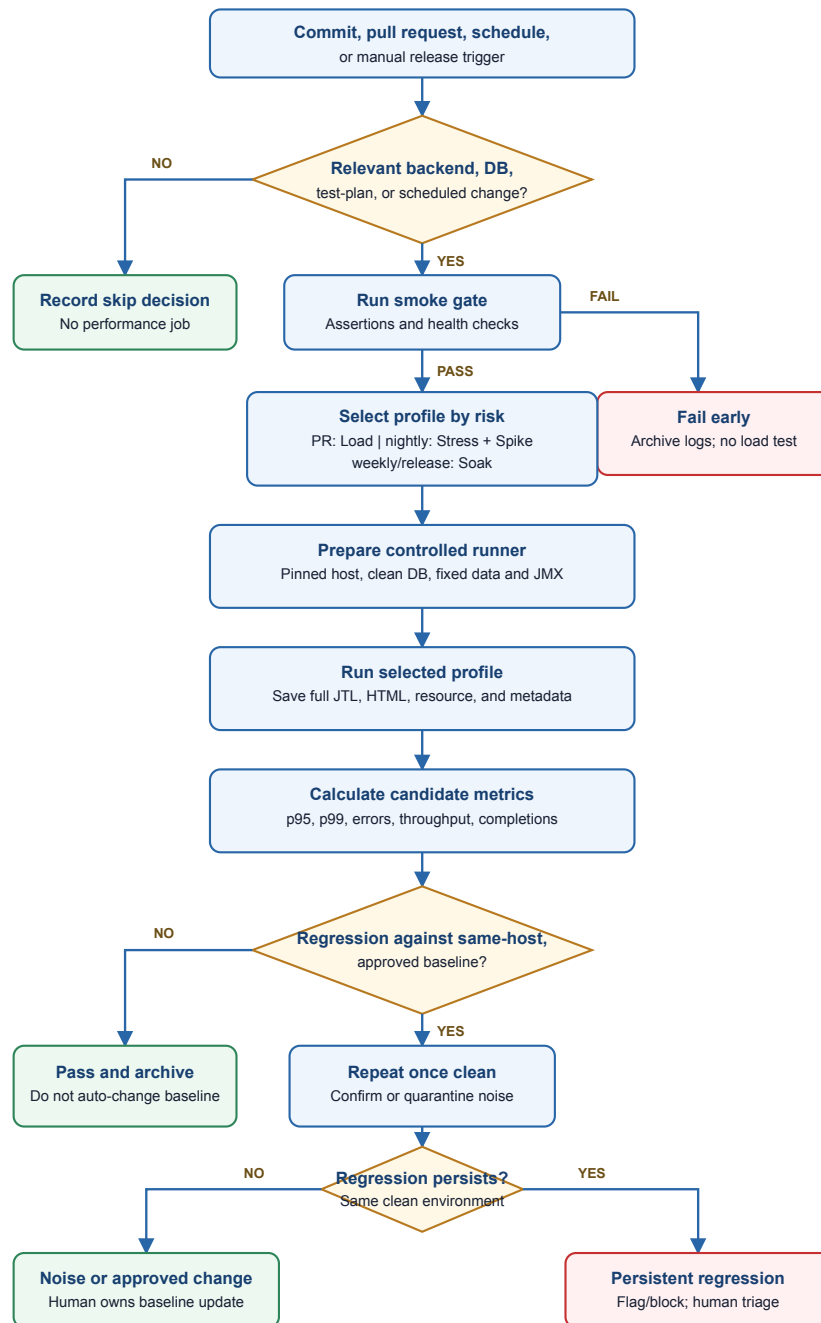
The latency bands represent +50%/+100% over the worst accepted p95 and p99. Throughput floors are 10% below observed values. Repeated clean runs must quantify natural variance before these proposed gates are activated.

12. Optimization judgment

Source review supports experiments with an explicit transaction around FR16 inserts, SQLite WAL plus `busy_timeout`, and a profiled non-unique `users(email)` index. It rejects a normal B-tree index as a direct fix for `LIKE '%term%'`, rejects an additional coupon-code index because `code` is already `UNIQUE`, and rejects a client/server-style connection pool as a drop-in fix for one SQLite handle. Search caching and clustered workers are unsupported by this unique-write/read-after-write workload and may add stale reads or write contention.

13. Task 3 - Continuous performance-testing model

The proposed pipeline watches commits but does not run every expensive profile on every change. Documentation/frontend-only changes record a skip decision. A relevant backend/auth/middleware pull request runs Smoke then Load; database/schema/SQL/import/dependency or test-data changes add Stress; nightly main runs add Stress and Spike; weekly and release-candidate runs add the 10-15 minute Soak. A maintainer may promote any change when path rules understate its risk.



Smoke must pass all HTTP and business assertions before performance traffic starts. The controlled runner then restores the canonical database, seeds the 80 accounts, and records the commit SHA, JMX/CSV hashes, hostname, CPU/RAM/OS, Node/JMeter versions, PID, and timestamps. Every job retains its full JTL, HTML report, backend log, resource CSV, database pre/post state, and run metadata. A failure therefore remains attributable to code, workload, data, and hardware rather than only to a dashboard number.

14. p95 regression and baseline ownership

The Task 2 thresholds are bootstrap candidate gates, not an SLA. Any assertion failure stops the pipeline. Stable-stage p95 warns above 7.5 ms and fails above 10 ms after confirmation; p99 warns/fails above 9/12 ms. Candidate throughput floors are 14.44 samples/s at Load 6 VUs, 55.99 at Stress 24 VUs, 75.63 during the Spike burst, and 71.54 samples/s or 17.89 workflows/s for Soak. Node CPU above 30% or RSS above 165 MB triggers investigation, not an automatic failure based on one peak.

Before enforcement, at least three clean runs on the pinned runner establish a median and natural spread. A candidate p95 regression requires both a relative increase of at least 20% and an absolute increase of at least 2 ms, or it crosses the approved 10 ms hard band. This dual test prevents a quantized move from 4 to 5 ms being called severe merely because it is 25%. The pipeline repeats a candidate once from a clean state. A persistent regression flags the commit and may block release; a non-repeating result is quarantined for human review.

Only a named human reviewer can approve a new versioned baseline and must record why the change is expected. The pipeline never automatically learns a slower run. Baselines are compared only when the JMX/data/database seed and environment fingerprint match; results from different hardware remain separate evidence.

15. Cost and false-alarm trade-offs

Trade-off	Risk	Control
Execution cost and feedback time	All profiles on every commit delay pull requests.	Smoke + Load on relevant PRs; Stress/Spike nightly; Soak weekly/release.
Noisy neighbours	Other processes alter CPU, I/O, and latency.	Dedicated/pinned host, reject contaminated runs, repeat once clean.
Data drift	A growing SQLite file or reused state changes query/write cost.	Restore the versioned seed and verify pre/post counts.
Small latency values	A 1 ms change can appear as a large percentage.	Require relative and absolute deltas plus confirmation.
Broad aggregates	Overall throughput can hide a slow sampler or partial journey.	Compare per-sampler percentiles and exact final FR06 completions.
Hardware/version drift	Unlike environments produce invalid comparisons.	Version the fingerprint and compare only like-for-like runs.
Baseline ownership	Auto-rebaselining normalizes regressions.	Require human approval and retain old/new artifacts.
Artifact storage	Full JTL/HTML/resource evidence grows over time.	Preserve failures/releases; apply retention to passing PR artifacts.

This model provides fast PR feedback, broader nightly coverage, and sustained release evidence while controlling cost and false alarms. The standalone proposal is retained in `task3_continuous_performance_proposal.md`.

16. Deliverables and conclusion

The primary `test-plans`, `results`, `reports`, and `evidence` paths now contain only the dated WF-04 baseline. Reproducible Task 1 and Task 2 metrics are in `analysis/`; the unreviewed AI first pass is preserved under `audit/`; the detailed correction is in `task2_analysis_review.md`; and the Task 3 proposal/flow chart are in `task3_continuous_performance_proposal.md` and `assets/`.

WF-04 is a coherent, source-grounded admin journey and is distinct from WF-01, WF-02, and WF-03 supplied by the student. All accepted JMeter/business assertions passed, product deltas match successful imports exactly, the ten-minute fixed-work soak is complete, and Task 3 defines how future commits would be gated without silently changing the baseline. Conclusions remain bounded to the tested source, data, timers, host, and isolated port 3001.